Monday POC Kit — Homelab Proof-of-Concept

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Material to bring into the CEO meeting to prove the v1 LSP telemetry architecture is already running as a working system at home. Two tracks: **live demo** (if connectivity and time allow) and **printed backup pack** (always have, hand over at the end regardless).

Part 1 — Live demo

Pre-meeting checklist (do the evening before)

- ☐ Open the three target URLs on the laptop, leave tabs loaded in order:
  - Tab 1: Uptime Kuma dashboard (`:3002`)
  - Tab 2: Grafana alerting list (`:3000/alerting/list`)
  - Tab 3: Terminal with `grafana-twilio-bridge.js` open in `less` or `bat`
- ☐ Confirm all 11 Uptime Kuma monitors green. If any are red, either fix them before Monday or use the red one as the demo incident.
- ☐ Verify Tailscale reachability from laptop: `tailscale status` shows `nixos` as online. If it doesn't, you have no demo — fix before Monday.
- ☐ Phone hotspot tested, laptop tethered, services reachable through it. Berrinba office WiFi is unknown; don't rely on it.
- ☐ Phone with SIM in it. The Twilio alerts land on the on-call number — put the phone screen-up on the table at the start.
- ☐ USB drive with PDF export of this entire pack (proposal + POC kit + diagrams) as a laptop-failure fallback.
- ☐ Printed pack (Part 2 below) in a folder, ready to hand over at the end of the meeting whether the live demo happened or not.

Demo script — 5 minutes, or shorter

Open the demo with the frame, not the tech. The tech is the proof — the frame is what the CEO remembers.

"This is the same stack I'm proposing for LSP v1, running in my home environment for about a month. Let me trace one alert end-to-end."

Tab 1 — Uptime Kuma (90 seconds)

- Point at the 11 monitors list, all green.
- *"These are the HTTP and TCP checks running every 60 seconds. Three consecutive failures fires an alert — around three minutes end-to-end."*
- Click one monitor (pick Immich or Plex — recognisable). Show the uptime graph.
- *"In LSP v1, each row here becomes an endpoint we care about at each site: a WiFi gateway, an LSPcare login page, a bed terminal."*

Tab 2 — Grafana Alerting (90 seconds)

- Show the alert rules page. 10 rules across Host Health, Systemd Units, Proxmox Health.
- Click into one with a clear story — "NAS Mount Gone" works well.
- *"This is the second layer. Prometheus scrapes metrics, Grafana evaluates rules, fires when conditions sustain. For LSP v1 this layer is deferred to v2 — too much setup per site for a detection-first design. But it's ready when we need post-incident investigation."*

Optional — trigger a live alert (90 seconds, only if the meeting is going well)

- Open a terminal, stop a non-critical service:

  ```
  systemctl --user stop sunshine.service
  ```

  or similar — pick something whose outage is visible in Uptime Kuma within 60 seconds.
- Keep talking while the alert cooks. Reference your phone face-up on the table.
- When the SMS lands, pick up the phone, read the alert aloud.
- *"That's the detection half of v1. The other half — the Zoho ticket — is the component I'd build on top."*
- Restart the service: `systemctl --user start sunshine.service`. Confirm the recovery SMS when it arrives.

Skip this step if you're running short on time or if the first two tabs have landed. The printed pack carries the point for you.

Tab 3 — The bridge code (60 seconds)

- Open `~/nixos-config/scripts/grafana-twilio-bridge.js` in the terminal.

- *“Forty lines of Node.js. Receives Grafana’s webhook JSON, posts to the Twilio API. Wrapped in a systemd service, credentials loaded from sops-nix — never in git.”*
- Scroll to the `LoadCredential` line if visible.
- *“For LSP v1, this same pattern plumbs the Zoho Desk integration alongside Twilio. One module, two downstreams.”*

Close (30 seconds)

> *“Nothing in the v1 proposal needs to be invented. It’s this stack, repointed at LSP endpoints, with Zoho Desk added as a second downstream next to Twilio. What I’m proposing to you is to productionise a pattern I already run and operate.”*

Known risks and fallbacks

Risk	Fallback
Berrinba WiFi won’t route to Tailscale	Phone hotspot
Phone hotspot not working	Skip to printed pack, do not improvise
Laptop battery dies / hard failure	Printed pack covers everything the demo shows
A service is genuinely down at demo time	**Use it.** Point at the fired alert, say “this is what you’d see at a customer site.”
Live-trigger alert takes >3 min	Keep talking through tabs 2 and 3; the SMS lands when it lands
Meeting runs short	Drop live demo entirely. Proposal + printed pack is sufficient

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Part 2 — Printed backup pack (leave-behind, 4 pages)

Print double-sided on good paper, staple top-left, hand to the CEO at the end of the meeting. He keeps this.

Page 1 — Homelab architecture today

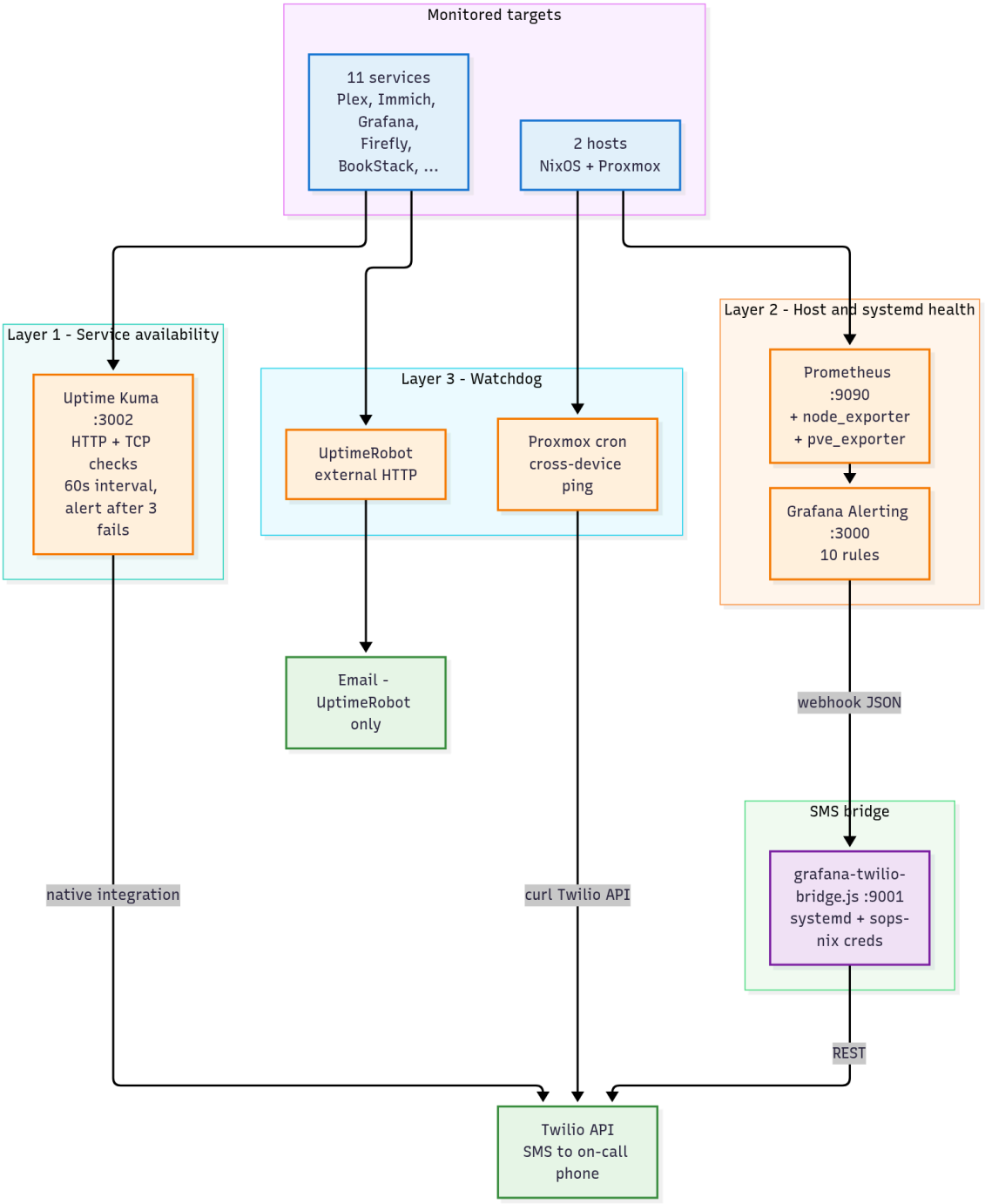


Figure 1: Homelab monitoring architecture

Caption: *The current homelab stack. Three detection layers, one SMS bridge, one Twilio downstream. For LSP v1 the Prometheus/Grafana layer is deferred to v2 and a Zoho Desk integration sits in parallel to Twilio on the output side.*

Page 2 — Alert flow, end to end

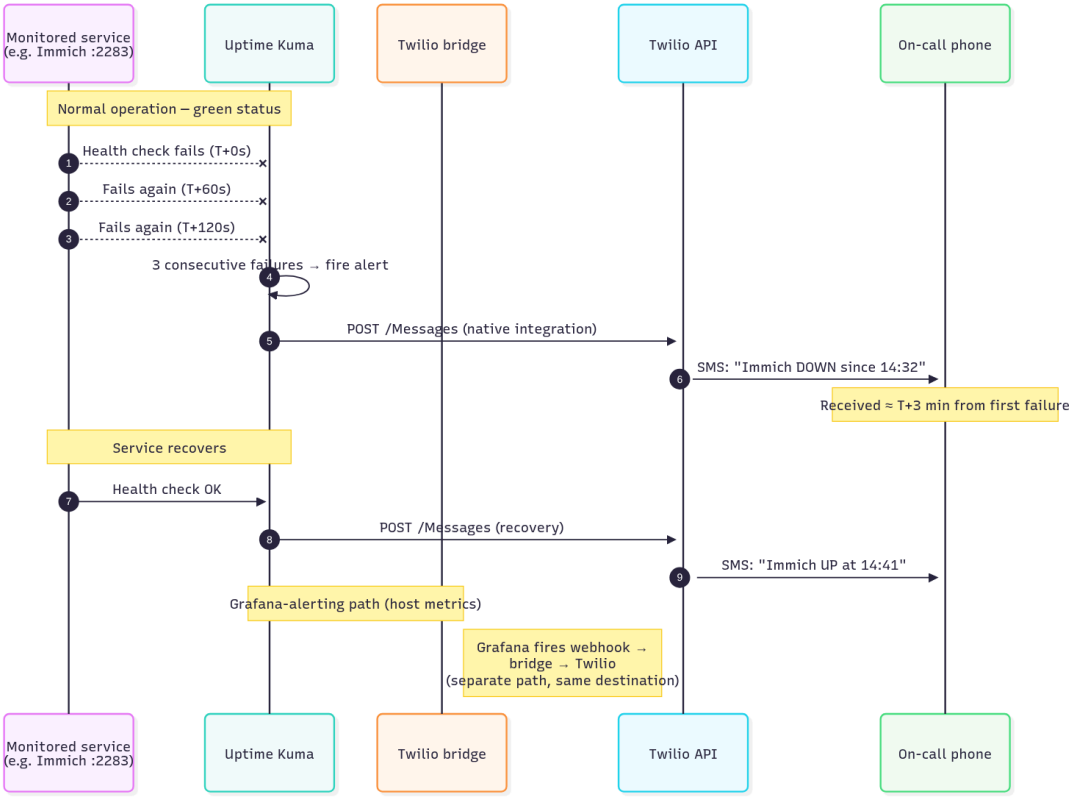


Figure 2: Alert flow sequence — service down to SMS in hand

Caption: *Full round-trip of a detected incident, from first failed health check to on-call SMS. Three minutes from failure to phone is the current configured budget; it's a dial we can tune per signal severity.*

Page 3 — Alert inventory

The 14 alert rules running today.

Layer 1 — Service availability (Uptime Kuma, :3002)

Target	Check type	Alert condition
Plex	HTTP	3 consecutive 60s failures
Audiobookshelf	HTTP	3 consecutive 60s failures
Immich	HTTP	3 consecutive 60s failures
Overseerr	HTTP	3 consecutive 60s failures
qBittorrent	HTTP	3 consecutive 60s failures
Grafana	HTTP	3 consecutive 60s failures
Prometheus	HTTP	3 consecutive 60s failures
Firefly III	HTTP	3 consecutive 60s failures
BookStack	HTTP	3 consecutive 60s failures
PostgreSQL	TCP	3 consecutive 60s failures
MariaDB	TCP	3 consecutive 60s failures

Layer 2 — Host and systemd health (Grafana Alerting)
NixOS server (6 rules)

Alert	Condition	Sustained
Disk Full	filesystem > 90%	5 min
NAS Mount Gone	/mnt/nas absent	3 min
CPU Temp High	Tctl > 85°C	5 min
NVMe Temp High	> 70°C	5 min
Cloudflared Down	systemd unit not active	2 min
Tailscaled Down	systemd unit not active	2 min

Proxmox host (4 rules)

Alert	Condition	Sustained
Proxmox Temp High	k10temp > 70°C	5 min
Proxmox Disk Full	filesystem > 90%	5 min
TrueNAS VM Down	pve_up{qemu/101} < 1	3 min
Proxmox Memory High	memory > 90%	5 min

Layer 3 — Watchdog

- **Proxmox cron script** — cross-device ping check every 3 min, alerts after 3 consecutive failures (~9 min).
- **UptimeRobot** — external HTTP watchdog on public hostname, 5 min interval, email alerts.

Page 4 — Mapping homelab to LSP v1

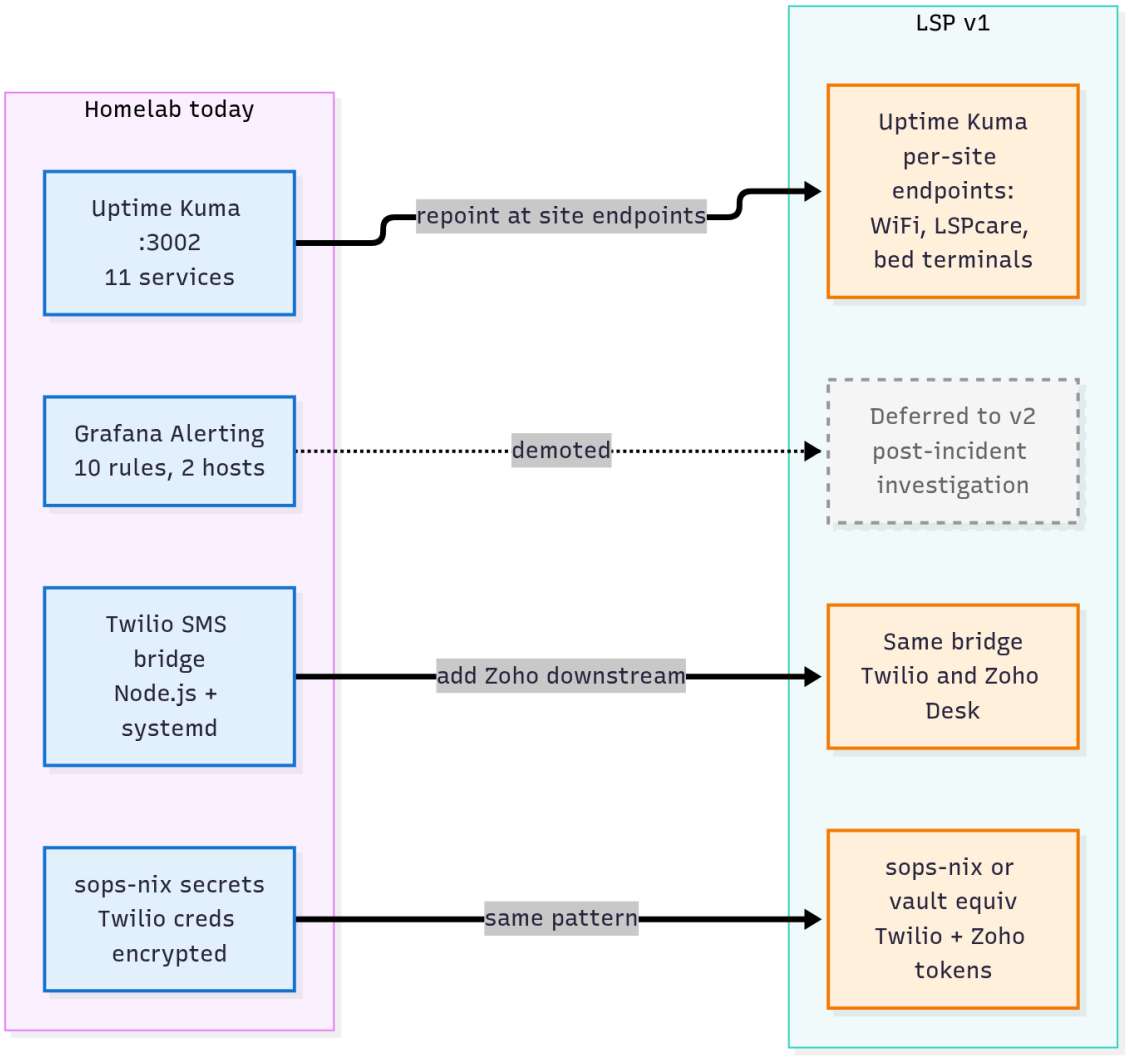


Figure 3: Homelab components mapped to their LSP v1 equivalents

What transfers and what doesn't:

Homelab today	LSP v1
Uptime Kuma, 11 services	Uptime Kuma, per-site endpoints (WiFi, LSPcare, bed terminals)
Grafana Alerting, 10 rules, 2 hosts	Deferred to v2 (post-incident investigation layer)
Twilio SMS bridge, Node.js, systemd	Same bridge, Twilio unchanged, Zoho Desk added in parallel
sops-nix secrets	sops-nix or vault equivalent — same pattern, never in git
One Twilio number, one on-call phone	Routed by severity and site, multiple on-call rotations if needed
One home, one Proxmox host	10–20 sites in pilot, extending toward 60+ in v2 and beyond

Bottom line: everything in LSP v1 is this stack, repointed and re-wrapped. The novel engineering is the Zoho Desk integration layer and the per-site onboarding flow — not the monitoring core.

Part 3 — Likely CEO questions, with Ivan's pre-prepared answers

Not scripted word-for-word — just the shape of the answer you want to give, so you don't have to improvise under pressure.

"Why not just extend what Saran started?"

Saran's design is excellent for an ops-engineer infrastructure platform. Your vision is for support staff to catch customer-visible problems before the customer. That's a different problem — the detection surface is user-experience endpoints, not infra metrics. So I'd keep Saran's design as the v2 investigation layer, but lead with endpoint probes for v1. It's not binned, just resequenced.

"How do I know you'll finish it?"

You don't, in the sense that no one can promise that. What you can look at: I've shipped the exact same pattern at home end-to-end — monitoring, alerting, SMS, infrastructure-as-code, secrets management, verified restore drills. I'd propose a monthly check-in so slippage is visible early, and v1 scope sized assuming 2–3 weeks of lost time per quarter from the Sales day job.

“What if the Sales workload spikes?”

v1 scope is sized assuming it happens. I’d rather commit to 10 sites in six months and deliver cleanly than commit to 30 and leave half of them half-wired. If quarterly Sales volume means I lose a month, the milestone list shifts by a month — I’d flag it at the check-in rather than silently slipping.

“Zoho Desk — are we locked in forever?”

No. The integration layer is pluggable by design. If you decide in 2027 that Freshdesk or Jira Service Management is a better fit, we reconfigure the integration, we don’t rebuild the system. v1 ships with Zoho because that’s what you use today.

“What’s the risk this becomes a side project that quietly dies?”

Three things reduce that risk: (1) the arrangement is written down with a milestone list, (2) monthly check-ins catch slippage early, (3) v1 is sized small enough that even a slow quarter still ships a working pilot site. If after six months we’ve landed three sites and it’s clearly not working, we end it cleanly — that’s a better outcome than drifting.

“Can we talk about portfolio rights later?”

I’d rather settle the shape now, even if we don’t sign anything today. It’s part of what makes this arrangement work for me — I keep the engineering output on my record. “I can describe the architecture publicly but not customer names, deal sizes, or confidential config” is a reasonable starting line.

“What about Jason?”

Jason’s networking background is complementary — he owns the network layer at sites, I own the monitoring layer on top of it. I’d want him in the loop operationally from week one, especially on the connectivity side (VPN, Tailscale, site routing). This isn’t me stepping around him.

“Can we start with just the AI receptionist and come back to telemetry?”

The two are compatible — the receptionist is a weeks-not-months engagement, and the telemetry build is paced in hours per week. I’d be fine sequencing them with the receptionist first if that fits better. Let’s agree a rough order and I’ll come back with scope on both.

“How do I keep track of this if I’m not in the detail?”

Monthly written check-in: current state, what shipped in the last month, what’s shipping next month, blockers. Fifteen minutes to read, in your inbox on the

1st. Plus I’d propose the status board itself is accessible to you and the support team from day one — you see live state whenever you want.

Part 4 — Post-meeting follow-up template

Send within 48 hours of the meeting while the alignment is fresh. Short, plain email.

```
Subject: LSP telemetry — follow-up from Monday

Hi [CEO name],

Thanks for the time on Monday. Capturing what I heard so we're
aligned before I come back with a scoped plan:

- Arrangement: I continue in Sales & Project Specialist, telemetry
  becomes a parallel engagement at current salary, ~10 hrs/week.
- Priority: customer-visible problem detection first. Infra metrics
  come in v2.
- v1 scope: endpoint probes, status board, Twilio SMS, Zoho Desk
  ticket integration.
- Portfolio rights: [what you agreed, or "to be defined before I
  start"].
- Next step from me: written scope + milestone plan by [date,
  within one week].

If any of the above looks wrong, let me know this week so we start
from the right place. Happy to walk through any of it on a call
before I write the scope doc.

Ivan
```

If the CEO said yes in principle but hasn’t committed: add a line at the top asking for confirmation on whether to proceed to scope-doc stage.

If the CEO said he wants to think about it: send the same email minus the “next step” line, with a soft re-raise date two weeks later in your own calendar.

If the CEO said no: still send a short thank-you. The relationship outlives the outcome.